1.6 Outer Measure

We want to define a set function m that maps a set $E \subseteq R$ to a value in R such that the following properties hold:

- Interval Length: for any interval I, its measure $m(I)$ is the difference between its end points
- Translation Invariant: for any set E, we have $m(E) = m(E + \lambda)$
- Countable Additivity: for any countable disjoint sets $E_1, \ldots, E_n, \ldots$, we have $m(E_1 + \ldots + E_n + \ldots) = m(E_1) + \ldots + m(E_n) + \ldots$

It turns out that such a set function cannot be defined on all sets in R. Instead, it is possible to define it on a $\sigma-$algebra of R.

In what follows, we first define an **outer measure** for all sets in R. Then from there we construct a measure function, called **Lebesgue measure**, that is defined on a $\sigma-$algebra.

Outer measure

For an open interval $I = (a, b)$, its length is denoted by $l(I) = b - a$. If $a = -\infty$ or $b = \infty$ or both, then $l(I) = \infty$.

For a set $E \subseteq R$, we define the following outer measure function $m^*(E) = \inf\{\sum_{k=1}^{\infty} m(I_k) \mid E \subseteq \bigcup_{k=1}^{\infty} I_k\}$, where the collection of countable open intervals $\{I_k\}_{k=1}^{\infty}$ is an open cover of E.

In other words, the outer measure of a set E is the infimum of the "total lengths" of its open covers.

The outer measure satisfies the properties of

- Interval Length
- Translation Invariant
- Monotonity
- Countable Sub-additivity . These properties are shown by the following propositions.

Proposition 1.6.1: If $A \subseteq B$, then $m^*(A) \leq m^*(B)$. *Proof:* Let $S(A) = \{\sum_{n=1}^{\infty} l(I_n) \mid A \subseteq \bigcup_{n=1}^{\infty} I_n\}$. $S(B)$ is defined similarly. Since $A \subseteq B$, any cover of B is also a cover of A. In other words, $S(B) \subseteq S(A)$. Therefore, $\inf\{S(A)\} \leq \inf\{S(B)\}$. **QED**

Proposition 1.6.2: For any countable set E, $m^*(E) = 0$. *Proof:* For any integer n, there is one-one correspondence between the countable sequence $\{a_{n,k} = \frac{1}{2^n} \cdot \frac{1}{2^k}\}_{k=1}^{\infty}$ and E. So $\{(x_k - a_{n,k}, x_k + a_{n,k})\}_{k=1}^{\infty}$ is an open cover of E, where $x_k \in E$. By definition, $(1) \quad m^*(E) \leq \sum_{k=1}^{\infty} a_{n,k} = \frac{1}{2^n}$. Since (1) holds for arbitrary large n, we must have $m^* E = 0$. **QED**

Proposition 1.6.3: The outer measure of the empty set is 0. That is, $m^*(\emptyset) = 0$. *Proof:* This is obvious, since any open interval $(-\frac{\epsilon}{2}, \frac{\epsilon}{2})$ is an open cover of $\emptyset$. Therefore, $m^*(\emptyset) \leq \epsilon$ for arbitrary small ϵ. **QED**

Proposition 1.6.4: For intervals $(a, b), (a, b], [a, b), [a, b]$, their outer measures are $b - a$. *Proof:* We first assume that $-\infty < a, b < \infty$. For the closed interval $I = [a, b]$, any $O_\epsilon = (a - \epsilon, b + \epsilon)$ is an open cover of I. Therefore, $m^*(I) \leq (b - a) + 2\epsilon$ for arbitrary small ϵ. As a result, we must have $m^*(I) \leq b - a$.

On the other hand, we would like to show $m^*(I) \geq b-a$. It suffices to show that for any $I \subseteq \bigcup I_k$, we have $\sum l(I_k) \geq b-a$. To this end, note that I is bounded and closed. By Heine-Borel Theorem, there is a finite sub-cover $I_1, \ldots, I_N$. Then there is some $I_{j_1} = (a_1, b_1)$ such that $a_1 < a < b_1$. If $b_1 \geq b$, we are done. Otherwise, $b_1 < b$ and there is some $I_{j_2} = (a_2, b_2)$ such that $a_2 < b_1 < b_2$. Again, if $b_2 \geq b$, we are done. Otherwise, there is some such I_{j_3}. We can easily prove by induction that $\sum_{p=1}^{M} l(I_{j_p}) \geq b_M - a$. Since $I_{j_1}, \ldots, I_{j_N}$ covers $[a,b]$, we must have $b_N > b$. Therefore, $\sum_{p=1}^{N} l(I_p) \geq b-a$. This completes the proof that $m^*([a,b]) = b-a$.

For intervals $(a,b), [a,b), (a,b]$, they are all subsets of $[a,b]$. By Proposition 1.6.1, we have $m^*(I) \leq b-a$. On the other hand, any closed intervals $[a+\epsilon, b-\epsilon] \subseteq I$. Thus, $m^*(I) \geq b-a-2\epsilon$. Since ϵ can be arbitrary small, we must have $m^*(I) \geq b-a$.

Finally, assume b is ∞. Since $(a,M) \subseteq (a,\infty)$, we have $m^*((a,\infty)) \geq M-a$ for arbitrary large M. Therefore, $m^*((a,\infty)) = \infty$. The case for $[a,\infty), (-\infty,b), (-\infty,b]$ can be shown similarly. **QED**

Proposition 1.6.5 (Translation Invariant): For any set E and any real number λ, $m^*(E) = m^*(E+\lambda)$. *Proof:* This is obvious, since for any interval I we have $l(I) = l(I+\lambda)$. **QED**

Proposition 1.6.6 (Countable Subadditivity): For any countable collection of sets $E_1, \ldots, E_n, \ldots$, disjoint or not, $m^*(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} m^*(E_n)$. *Proof:* Let $S(E_n)$ denote the countable collection of open intervals that cover E. So $l(S(E_n)) \leq m^*(E_n) + \frac{\epsilon}{2^n}$. Note that the collection $S(E_1), \ldots, S(E_n), \ldots$ is also a cover of $\bigcup E_n$. Therefore, $(1) \quad m^*(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} l(S(E_n)) \leq [\sum_{n=1}^{\infty} m^*(E_n)] + \epsilon$. Note that (1) holds for arbitrary small ϵ. Thus the conclusion follows. **QED**

Exercises

Q1. Let m be a function that is defined on a $\sigma-$algebra A, and takes value in $[0,\infty]$. Assume that m is additive over any countable disjoint collection of sets in A.

- (a) If $E, F \in A$ and $E \subseteq F$, then $m(E) \leq m(F)$.
- (b) If there exists E such that $m(E) < \infty$, then $m(\emptyset) = 0$.
- (c) For any countable collection of sets $E_1, \ldots, E_n, \ldots \in A$, we have $m(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} m(E_n)$.

Solution: (a) Since $E, F-E$ are disjoint and $E \cup F - E = F$, by the property of m we have $m(F) = m(E) + m(F-E)$. Since $m(F-E) \geq 0$, we have $m(E) \leq m(F)$. Note that it is not difficult to show $F - E \in A$.

(b) Note that $E, \emptyset$ are disjoint, and $E \cup \emptyset = E$. Thus, $m(E) = m(E) + m(\emptyset)$. Since $m(E) < \infty$, we have $m(\emptyset) = 0$.

(c) Take $F_1 = E_1$, and $F_n = \bigcup_{k=1}^{n} E_k - \bigcup_{k=1}^{n-1} E_k$ for $n \geq 2$. Clearly, every $F_n \in A$. Furthermore, $\{F_n\}$ are mutually disjoint and $\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} Fn$. Thus, by the additivity of m we have $m(\bigcup_{n=1}^{\infty} E_n) = m(\bigcup_{n=1}^{\infty} F_n) = \sum_{n=1}^{\infty} m(F_n)$. Finally, note that $m(F_n) \leq m(E_n)$ by the result of (a). Thus, we have the conclusion as desired. **QED**

Q2. (Counting Measure) A set function c is defined on all subsets of R. If a set E is infinite, then $c(E) = \infty$. If E is finite, then $c(E)$ is the number of elements in E. Show that c is countably additive and translation invariant. *Solution:* Clearly, c is translation invariant. For countable additivity, consider a countable disjoint collection of sets $E_1, \ldots, E_n, \ldots$. If there exists at least one infinite E_n, then the conclusion holds. So we may assume $E_1, \ldots, E_n, \ldots$ are all finite. Clearly, the total number of elements in $\bigcup_{n=1}^{\infty} E_n$ is equal to the sum of number of elements in $E_1, \ldots, E_n, \ldots$. **QED**

Q3. By using the method of outer measure, prove that $I = [0,1]$ is uncountable. *Solution:* By Proposition 1.6.4, $m^*(I) = 1$. If I is countable, then by Proposition 1.6.2 $m^*(I) = 0$, which is a contradiction. **QED**

Q4. Let A be the set of irrational number in $I = [0, 1]$. Prove that $m^*(A) = 1$. *Solution:* Since $A \subseteq I$, we have $m^*(A) \leq m^*(I) = 1$. On the other hand, since $I = A \cup I - A$, we have $m^*(I) \leq m^*(A) + m^*(I - A)$. Note that $I - A$ is the set of rational number and thus countable. So $m^*(I - A) = 0$. Therefore, $m^*(A) \geq m^*(I)$. **QED**

Q5. A set G is said to be a G_δ set provided that it is the intersection of a countable collection of open sets. Show that for any set E, there is a G_δ set G such that $E \subseteq G$ and $m^*(E) = m^*(G)$. *Solution:* By the definition of outer measure, for any n there exists an open set O_n such that $E \subseteq O_n$ and $m^*(O_n) < m^*(E) + \frac{1}{n}$. Let $G = \bigcap_{n=1}^{\infty} O_n$. Then G is a G_δ set, and $G \subseteq O_n$ for any n. It is not difficult to show that $E \subseteq G$. Thus, $m^*(E) \leq m^*(G)$. On the other hand, since $G \subseteq O_n$ for any n, we have $m^*(G) \leq m^*(O_n) < m^*(E) + \frac{1}{n}$. Therefore, we must have $m^*(G) \leq m^*(E)$. **QED**

Q6. Let B be the set of rational numbers in $[0, 1]$. Let $\{I_k\}_{k=1}^{\infty}$ be a finite collection of open intervals that cover B. Show that $\sum_{k=1}^{n} m^*(I_k) \geq 1$. *Solution:* It is not difficult to show that there are at most n irrational numbers $p_1, \ldots, p_n \in [0, 1]$ that are not covered by $I_1, \ldots, I_n$. So let P be the set of $p_1, \ldots, p_n$. Clearly $m^*(P) = 0$. Note $m^*([0, 1] - P) + m^*(P) \geq m^*([0, 1]) = 1$. Thus, $\sum_{k=1}^{n} m^*(I_k) \geq m^*([0, 1] - P) \geq 1$. **QED**

Q7. Prove that if $m^*(A) = 0$, then $m^*(A \cup B) = m^*(B)$. *Solution:* Since $B \subseteq A \cup B$, we have $m^*(B) \leq m^*(A \cup B)$. On the other hand, since $m^*(A \cup B) \leq m^*(A) + m^*(B)$ and $m^*(A) = 0$, we have $m^*(A \cup B) \leq m^*(B)$. **QED**

Q8. Let A, B be bounded sets for which there exists an $\alpha > 0$ such that $|x - y| \geq \alpha$ for any $x \in A, y \in B$. Show that $m^*(A \cup B) = m^*(A) + m^*(B)$. *Solution:* Let $r = \frac{\alpha}{4}$. Then $\{(x - r, x + r)\}_{x \in A}$ is an open cover of A. And the open set $O_A = \bigcup_{x \in A}(x - r, x + r)$ is a countable union of a disjoint collection of open intervals. Clearly, $A \subseteq O_A$ and $B \cap O_A = \emptyset$. Similarly, we can construct an open set O_B such that $B \subseteq O_B$ and $A \cap O_B = \emptyset$.

Note that $O = O_A \cup O_B$ is an open cover of $A \cup B$. To prove $m^*(A \cup B) \geq m^*(A) + m^*(B)$, it suffices to show that for any open cover O' of $A \cup B$ such that $O' \subseteq O$, we have

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(1) \ \ \ m^*(O') \geq m^*(A) + m^*(B)
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